

Year 10 Science
Chemical Sciences
Practical - Precipitation Reactions

Aim: To observe precipitation reactions and determine the solubility of ions.

Method:

INVESTIGATION 5.1

Will it precipitate?

AIM To predict and test for precipitation when a variety of solutions are added to each other

Materials:

5 semi-micro test tubes and a test-tube rack
a white tile and a black tile
safety glasses
dropping bottles of the following solutions: copper sulfate, sodium chloride, silver nitrate, cobalt chloride, sodium hydroxide, potassium iodide

CAUTION Wear safety glasses.

METHOD AND RESULTS

- ▶ Place 10 drops of copper sulfate solution in each test tube.
 - ▶ Add 10 drops of sodium chloride to the first test tube, 10 drops of silver nitrate to the second, and so on until each tube contains copper sulfate solution and one other solution. Hold a black or white tile behind the test tube if necessary to detect the presence of a precipitate.
- 1 If there is a visible reaction, record your observations in a table.
 - ▶ Tip the residues into a waste bottle. Wash out the test tubes thoroughly and this time place 10 drops of sodium chloride in each of the test tubes. Again add one of the other solutions to each test tube (but not copper sulfate, which has already been tested).
- 2 Record your observations in your table.
 - ▶ Repeat until all possible pairs of solutions have been tested.

DISCUSS AND EXPLAIN

- 3 Write word equations for each of the pairs that reacted to form a precipitate.
- 4 Use formulae to write chemical equations for each of the pairs that reacted to form a precipitate.
- 5 You could have predicted which pairs of solutions would form a precipitate using the box of solubility rules under the heading *Soluble or not?* in this section. Check to see if the rules match your results.

- You will be supplied with a grid on which you can test for reactions. The background is black to make it easier to observe any reactions.
- Place 1 drop of each reactant on a square and record any reaction - usually a solid coloured substance will form. Record this with a tick ✓ and the colour.
- If there is no solid produced, record this with a X.

Observations

Negative ions Positive ions	NaCl	NaOH	NaBr	Na ₂ SO ₄	Na ₂ CO ₃	NaNO ₃
Ca(NO ₃) ₂						
Cu(NO ₃) ₂						
Mg(NO ₃) ₂						
KNO ₃						
AgNO ₃						
FeCl ₃						
Pb(NO ₃) ₂						

Discussion

- Do questions 3-5 above.

Conclusion: